Branch/Class/Batch: AI&DS/TE/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

EXPERIMENT NUMBER: 9

AIM

To configure an FTP server and transfer files between a client and the server to demonstrate the working of the File Transfer Protocol (FTP).

OBJECTIVES

1. To understand the concept and working of FTP.
2. To configure an FTP server and client.
3. To establish a connection between the client and the FTP server.
4. To transfer files between the client and server.

THEORY

File Transfer Protocol (FTP) is a standard network protocol used to transfer files between a client and a server over a TCP/IP network. The FTP server provides access to files, while the FTP client connects to the server to upload or download files.

FTP commonly uses TCP port 21 for control connections. Traditional FTP transfers data without encryption; therefore, secure alternatives such as SFTP or FTPS are preferred when security is required.

RESOURCES REQUIRED

1. Computer or laptop.
2. Windows/Linux operating system or network simulation software, such as Cisco Packet Tracer.

PROCEDURE

1. Open the required network simulation software Cisco packet Tracer.
2. Configure the IP address and subnet mask of the server and client systems.
3. Enable or install the FTP server service.
4. Create a user account and configure the required file-access permissions.
5. Verify connectivity between the client and server using the `ping` command.
6. Open the FTP client and connect to the server using its IP address.
7. Enter the required login credentials.

Branch/Class/Batch: AI&DS/TE/A1
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8. Use FTP commands such as `dir`, `get`, and `put` to list, download, and upload files, respectively.

9. Verify that the transferred files are available at the destination.

OBSERVATION

1. The client successfully establishes a connection with the FTP server.

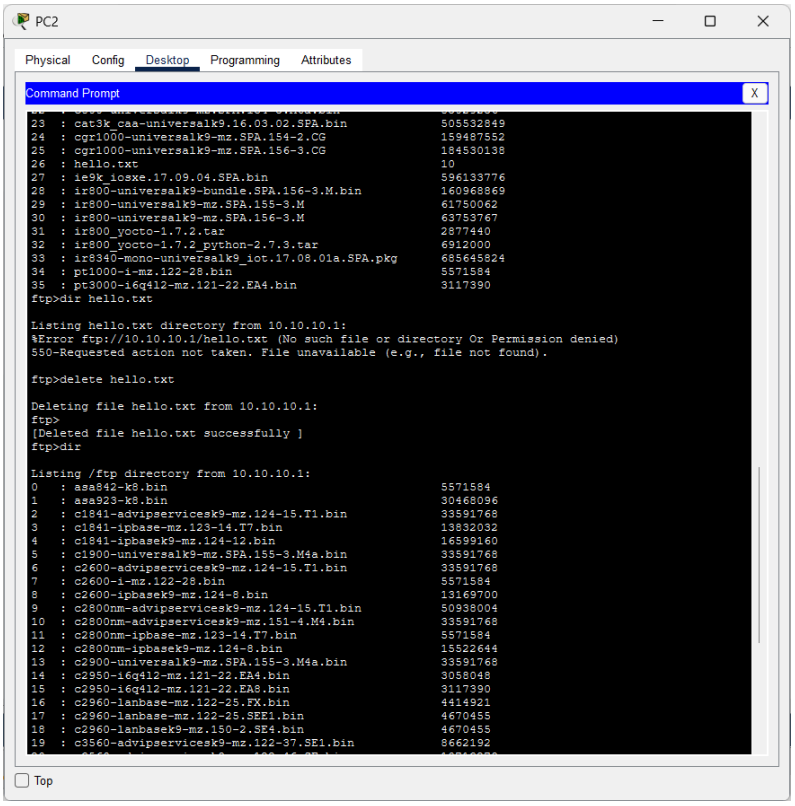
2. The server authenticates the user using the configured credentials.

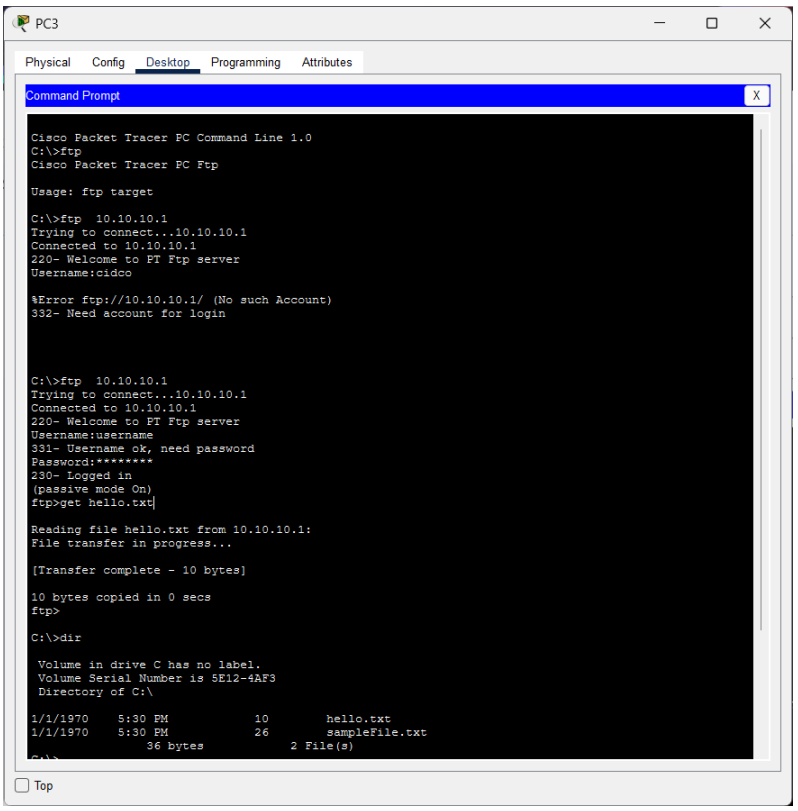
3. Files are listed and transferred between the client and server.

4. The transferred files are verified at the destination.

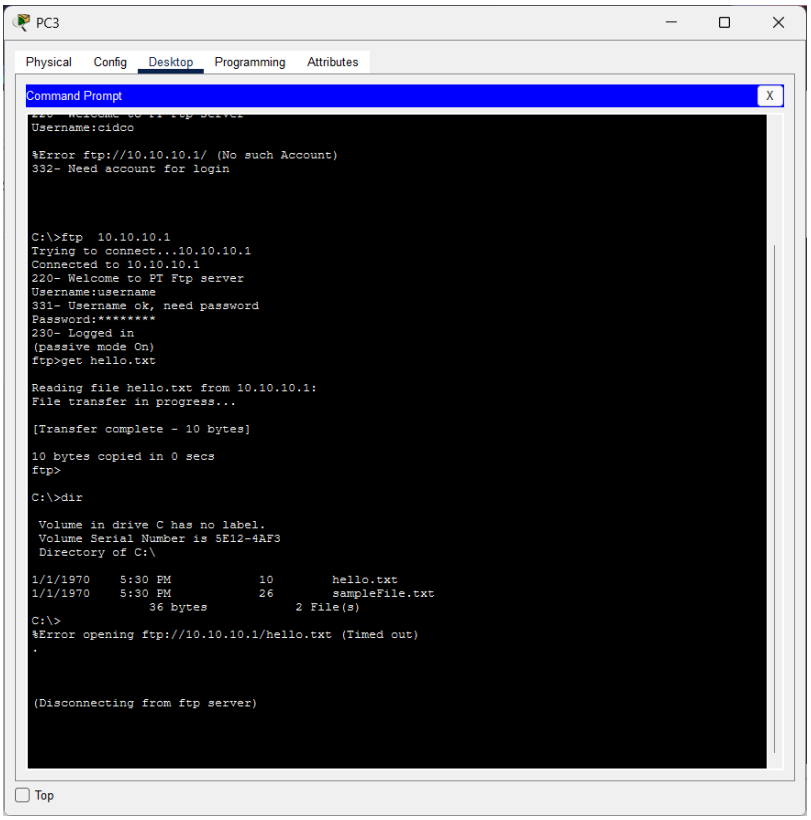
RESULT

Branch/Class/Batch: AI&DS/TE/A1
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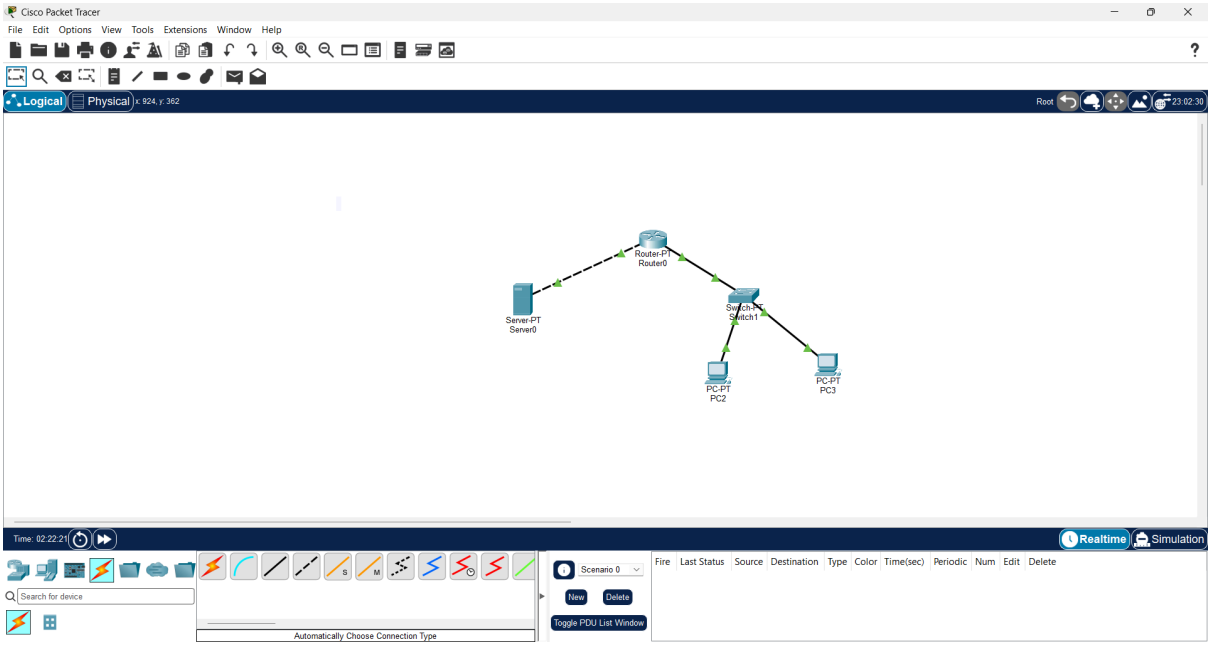


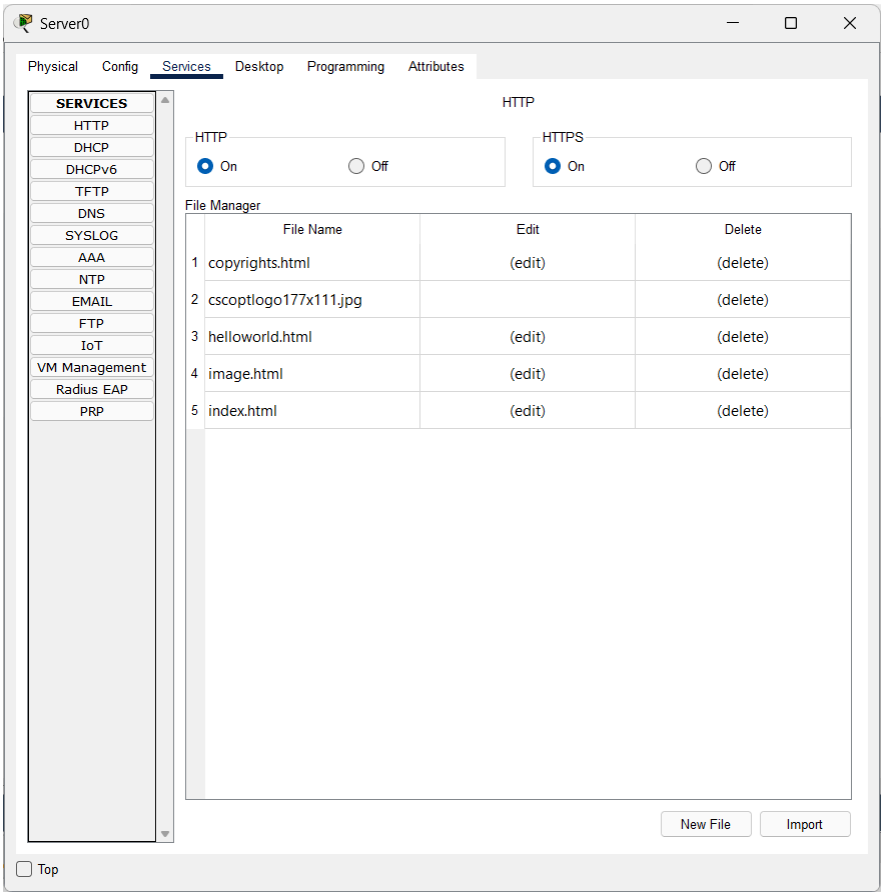


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CONCLUSION

An FTP server and client were successfully configured and verified using Cisco Packet Tracer. By authenticating over TCP port 21 and executing core FTP commands (dir, put, get), file transfers between the client and server were successfully demonstrated, confirming the fundamental working of the File Transfer Protocol.